Exercise 1:

- Set up tools (Python, Anaconda…etc.) or use online notebooks.

Exercise 2: getting started with Numpy.

- Run at least 2 tasks from each category of the next:

Array Creation and Manipulation:

- Create a 1D array with values ranging from 0 to 9.
- Create a 2D array with shape (3, 4) filled with zeros.
- Create an array with 10 equally spaced values between 0 and 1.
- Create a random 3x3 matrix with values between 0 and 1.
- Create an array with all even integers between 10 and 50.
- Create a 5x5 identity matrix.
- Create an array with 20 logarithmically spaced values between 1 and 100.
- Reshape a 1D array into a 2D array of shape (2, 5).
- Transpose a 2D array.
- Extract all odd numbers from an array using boolean indexing.

Array Operations:

- Add two matrices element-wise.
- Multiply two matrices using matrix multiplication.
- Calculate the mean, median, and standard deviation of an array.
- Find the sum of each column in a 2D array.
- Calculate the dot product of two 1D arrays.
- Normalize the values in an array between 0 and 1.
- Find the index of the maximum value in an array.
- Replace all negative values in an array with zeros.
- Calculate the element-wise product of two arrays.
- Calculate the element-wise square root of an array.

Indexing and Slicing:

- Slice a 2D array to extract the second column.
- Get the diagonal elements of a 2D array.
- Select rows from a 2D array where the sum of elements is greater than a threshold.
- Reverse the rows of a 2D array.
- Get the elements at odd indices of a 1D array.

Random Number Generation:

- Generate an array of 10 random integers between 1 and 100.

- Create a random 5x5 matrix with values from a normal distribution.
- Shuffle the elements of an array randomly.
- Generate an array of random integers following a Poisson distribution.
- Generate an array of random floats between 0 and 1 with a specified shape.

Boolean Indexing and Filtering:

- Create a new array containing only the positive values of an existing array.
- Find the elements in an array that are divisible by 3.
- Create a mask for elements greater than a specific value in an array.
- Replace all elements greater than 10 in an array with 10.
- Count the number of elements greater than 5 in a 2D array.

Statistical Operations:

- Calculate the correlation coefficient between two arrays.
- Find the index of the first occurrence of a specific value in an array.
- Calculate the cumulative sum of a 1D array.
- Calculate the 75th percentile of an array.
- Calculate the mean along a specific axis of a 2D array.

Broadcasting and Vectorization:

- Multiply each element in an array by a scalar.
- Add a 1D array to each row of a 2D array using broadcasting.
- Multiply corresponding elements of two arrays with different shapes using broadcasting.
- Calculate the outer product of two arrays.
- Apply a custom function element-wise to an array.

File I/O:

- Save a NumPy array to a text file.
- Load a NumPy array from a text file.
- Save a NumPy array to a binary file.
- Load a NumPy array from a binary file.
- Save multiple arrays to a single compressed NPZ file.

Exercise 3:

- V= [0.5, 0.8, 0.9]
- M = [1.0, 0.3, 0.65; 0.5, 0.8, 0.9; 1.0, 0.3, 0.65]

Using NumPy:

Create the vector V.

- Create the matrix M, M transpose, Matrix G = MT*M.
- Extract vector 1, row 1, sub matrix (0-2,0-2), then the max, min, sum, and average of each.
- Extract from M: (a) diagonal matrix. (b) identity matrix, (c) values bigger than 0.5.
- Insert a new row to M, and then insert a new column to M using random values.
- Calculate Euclidean distance of each two vectors of M using G.
- Verify if V is in M.
- Create a random sample of data (X values, and Y values) as to have 10 values of X in the range [0; 11], and Y has a linear form Y= aX + b (a in [1.5; 1.8], and b in [4; 4.5] (Check **Linspace).**
- Check if matrix N is a sub matrix of M.
- Create matrix of size (5,5) containing: (a) ones, (b) zeros, and (c) random values.

Exercise 4

Calculate Maclaurin series with NumPy

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \dots$$